

Pizza Sales Analysis

MySQL Workbench | SQL | Data Analysis Project

Project Introduction

This project analyzes pizza sales data using SQL in MySQL Workbench. A relational database was created by importing four CSV files and establishing relationships between them. SQL queries were used to answer business questions related to sales performance, customer ordering patterns, and revenue analysis.

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Database Schema & Tables

The PizzaBase database consists of four related tables that store order, order details, pizza, and pizza type information for sales analysis.

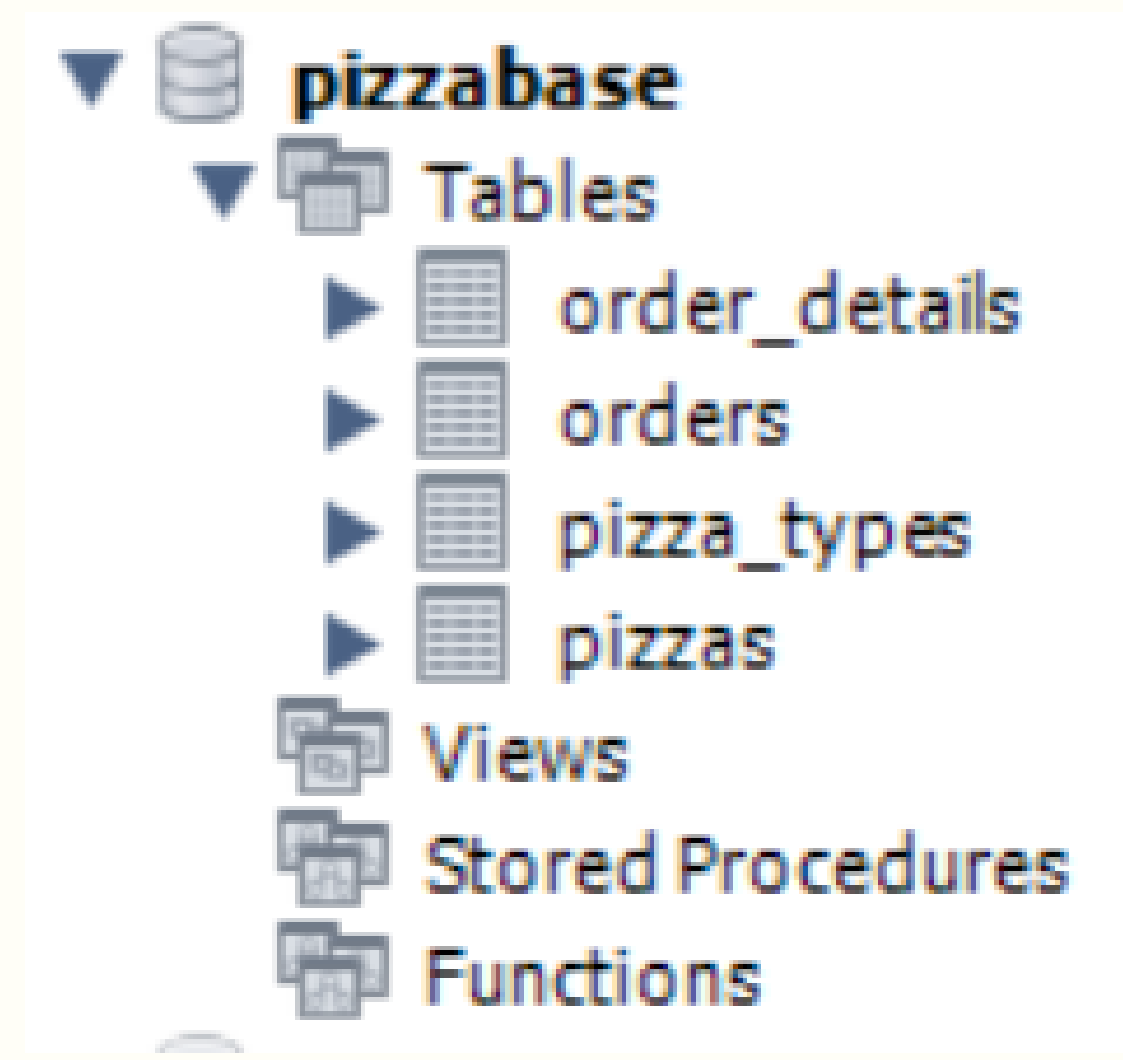


Table: order_details

Columns:

<u>order_details id</u>	int
	PK
order_id	int
pizza_id	tes
quantity	int

Table: orders

Columns:

<u>order_id</u>	int PK
order_date	date
order_time	time

Table: pizza_types

Columns:

pizza_type_id	text
name	text
category	text
ingredients	text

Table: pizzas

Columns:

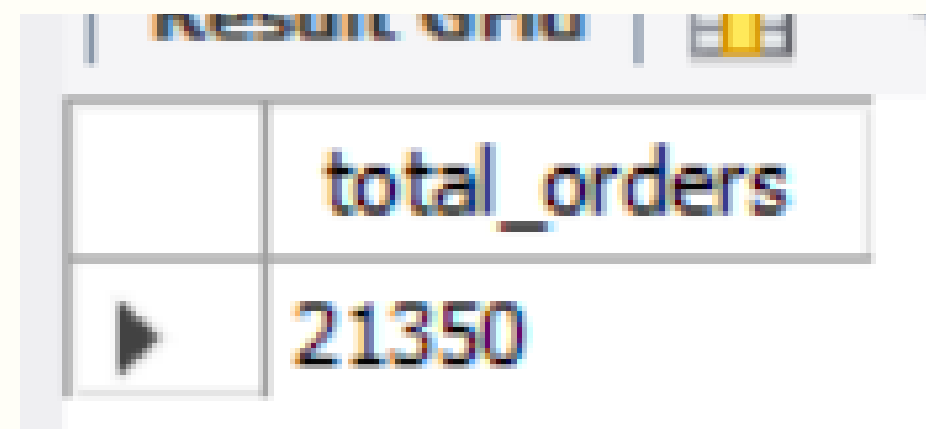
pizza_id	text
pizza_type_id	text
size	text
price	double

Order Analysis

- Retrieve the total number of orders placed.
- Determine the hourly distribution of orders.

```
-- QUE 1
-- Retrieve the total number of orders placed.

select count(order_id) as total_orders from orders;
```



	total_orders
▶	21350

Key Insight:

A total of 21,350 orders were placed during the recorded period.

```
-- QUE 7
-- Determine the distribution of orders
-- by hour of the day.
```

```
select hour(order_time) as hour,
count(order_id) as order_count
from orders
group by hour(order_time);
```

Key Insight:

Orders peaked during the 12 PM-1 PM time period.

hour	order_count
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642
21	1198
22	663
23	28
10	8
9	1

```
-- QUE 9
-- Group the orders by date and calculate
-- the average number of pizzas ordered per day.

select round(avg(quantity),0) as avg_pizza_per_day from
(select orders.order_date, sum(order_details.quantity) as quantity
from orders join order_details
on orders.order_id = order_details.order_id
group by orders.order_date) as order_quantity;
```

avg_pizza_per_day
138

Key Insight:

An average of 138 pizzas were ordered per day.

Product Analysis

- Highest-priced pizza.
- Most common pizza size.
- Top 5 most ordered pizza types.
- Quantity ordered by pizza category.

```
-- QUE 3 : Identify the highest-priced pizza.  
  
select pt.name, pz.price  
from pizza_types pt  
join pizzas pz  
on pz.pizza_type_id = pt.pizza_type_id  
order by pz.price desc  
limit 1;
```

name	price
The Greek Pizza	35.95

Key Insight:

The Greek Pizza was the highest-priced pizza at \$35.95.

```
-- QUE 4 : Identify the most common pizza size ordered.
```

```
select pizzas.size,  
count(order_details.order_details_id) as order_count  
from pizzas join order_details  
on pizzas.pizza_id = order_details.pizza_id  
group by pizzas.size order by order_count desc;
```

size	order_count
L	18526
M	15385
S	14137
XL	544
XXL	28

Key Insight:

Large (L) was the most ordered pizza size with 18,526 orders.



```
-- QUE 5 : List the top 5 most ordered pizza types
-- along with their quantities.

select pizza_types.name,
sum(order_details.quantity) as quantity
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.name
order by quantity desc
limit 5;
```

name	quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

Key Insight:

The Classic Deluxe Pizza was the most ordered pizza type with 2,453 orders.

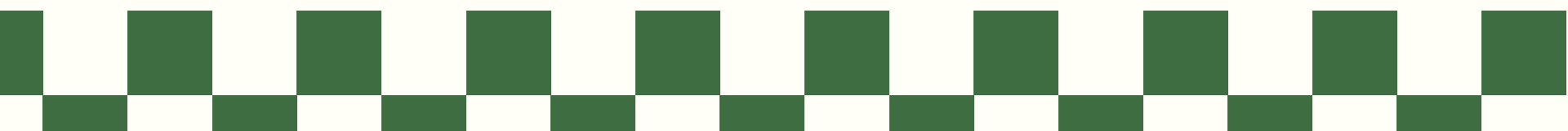

```
-- QUE 6 : Join the necessary tables to
-- find the total quantity of each pizza category ordered.

select pizza_types.category,
sum(order_details.quantity) as quantity
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on pizzas.pizza_id = order_details.pizza_id
group by pizza_types.category
order by quantity desc;
```

category	quantity
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

Key Insight:

The Classic category had the highest quantity ordered (14,888 pizzas).



```
-- QUE 8 : Join relevant tables  
-- to find the category-wise distribution of pizzas.
```

```
SELECT pt.category, COUNT(p.pizza_id) AS total_pizzas  
FROM pizza_types pt JOIN pizzas p  
ON pt.pizza_type_id = p.pizza_type_id  
GROUP BY pt.category ORDER BY total_pizzas DESC;
```

category	total_pizzas
Veggie	27
Classic	26
Supreme	25
Chicken	18

Key Insight:

Veggie has the highest number of pizza variants, followed by Classic and Supreme.

Revenue Analysis

- Calculate the total revenue generated.
- Determine the top 3 revenue-generating pizza types.
- Calculate the percentage contribution of each category.

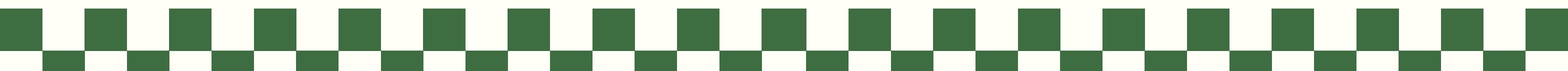
```
-- QUE 2
-- Calculate the total revenue generated from pizza sales.

select round(sum(d.quantity*p.price),2) as total_sales
from order_details d
join pizzas p
on p.pizza_id = d.pizza_id;
```

total_sales
817860.05

Key Insight:

Total revenue generated from pizza sales was \$817,860.05.



```
-- QUE 10 : Determine the top 3 most ordered
-- pizza types based on revenue.

select pizza_types.name,
sum(order_details.quantity * pizzas.price) as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.name order by revenue desc limit 3;
```

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

Key Insight:

The Thai Chicken Pizza generated the highest revenue.

```
-- QUE 11
-- Calculate the percentage contribution of each pizza type
-- to total revenue.

select pizza_types.category ,
round((sum(order_details.quantity * pizzas.price) /
(select round(sum(order_details.quantity * pizzas.price),2) as total_sales
from order_details join pizzas
on pizzas.pizza_id = order_details.pizza_id)* 100),2) as revenue_percentage
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category order by revenue_percentage desc;
```

category	revenue_percentage
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

Key Insight:

The Classic category contributed the highest share of total revenue (26.91%).

-- QUE 12 : Analyze the cumulative revenue generated over time.

```
select order_date,  
sum(revenue) over (order by order_date) as cum_revenue  
from  
(select orders.order_date,  
sum(order_details.quantity * pizzas.price) as revenue  
from order_details join pizzas  
on order_details.pizza_id = pizzas.pizza_id  
join orders on orders.order_id = order_details.order_id  
group by orders.order_date) as sales_daily;
```

order_date	cum_revenue
2015-01-01	2713.8500000000004
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05
2015-01-09	21526.4
2015-01-10	23990.350000000002
2015-01-11	25862.65
2015-01-12	27781.7
2015-01-13	29831.300000000003
2015-01-14	32358.700000000004
2015-01-15	34343.500000000001
2015-01-16	36937.650000000001
2015-01-17	39001.750000000001

colt 15

Key Insight:

Cumulative revenue increased steadily throughout the year, reaching \$817,860.05 by the end of the period.

```
-- QUE 13 : Determine the top 3 most ordered pizza types
-- based on revenue for each pizza category.

select category, name, revenue from
(select category, name, revenue,
rank() over(partition by category order by revenue desc) as rn
from
(select pizza_types.category, pizza_types.name,
sum((order_details.quantity)*(pizzas.price)) as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category, pizza_types.name) as a) as b
where rn <= 3;
```

category	name	revenue
Chicken	The Thai Chicken Pizza	43434.25
Chicken	The Barbecue Chicken Pizza	42768
Chicken	The California Chicken Pizza	41409.5
Classic	The Classic Deluxe Pizza	38180.5
Classic	The Hawaiian Pizza	32273.25
Classic	The Pepperoni Pizza	30161.75
Supreme	The Spicy Italian Pizza	34831.25
Supreme	The Italian Supreme Pizza	33476.75
Supreme	The Sicilian Pizza	30940.5
Veggie	The Four Cheese Pizza	32265.700000000065
Veggie	The Mexicana Pizza	26780.75
Veggie	The Five Cheese Pizza	26066.5

Key Insight:

The top revenue-generating pizza types were identified for each pizza category.

Project Summary

This project analyzed the sales performance of a pizza business using a dataset of 21,350 orders. The analysis revealed a total revenue of \$817,860.05, with Large pizzas being the most preferred size and the Classic category recording the highest quantity ordered. Customer demand peaked during the 12 PM–1 PM time period, while The Thai Chicken Pizza emerged as the highest revenue-generating pizza. Overall, the analysis provided valuable insights into customer preferences, product performance, and overall sales trends.